

Exam on Foundations of Neuro-Symbolic AI  
July 20, 2026  
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Matrikel number: \_\_\_\_\_

**Grading policy**

Choose the right answers to the questions. There is at least one right answer to each question, and some questions have multiple right answers.

- For each right answer you will get one point.
- For each wrong answer you will lose one point.
- Each question will be graded with at least 0 points.

You are allowed to bring one handwritten A4 sheet (both pages).

1. Let us consider the propositional formula:

$$f[X_0, X_1, X_2] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} [X_0, X_1, X_2]$$

How many coordinates does this formula have?

☐ 4    ☐ 16    ☒ 8    ☐ 3

What is the order of this formula?

☒ 3    ☐ 8    ☐ 2    ☐ 16

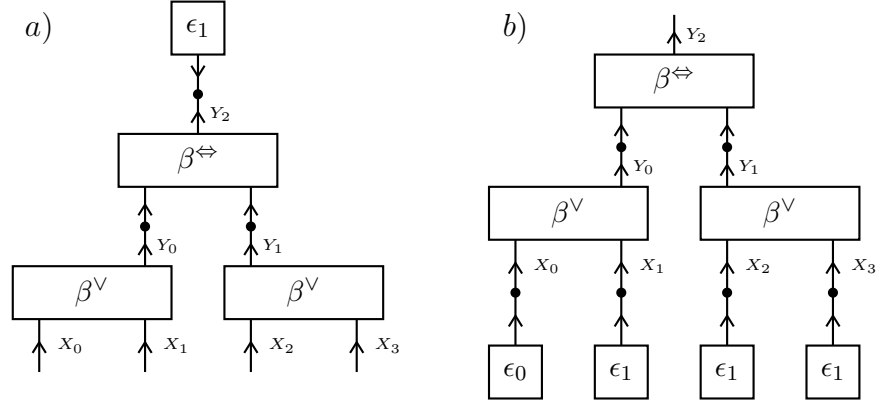
2. How many coordinates does a tensor of order 3 and leg dimensions  $m_0, m_1, m_2$  have?

☐  $m_0 \cdot m_2 + m_1 \cdot m_2$     ☐  $(m_0 \cdot m_1 \cdot m_2)^{m_0+m_1+m_2}$     ☒  $m_0 \cdot m_1 \cdot m_2$     ☐  $m_0 + m_1 + m_2$

3. Which of these is a syntactical representation of the formula in question 1?

☐  $X_0 \Rightarrow X_1$     ☐  $X_0 \Leftrightarrow (X_1 \vee X_2)$     ☐  $X_2 \Rightarrow X_1$     ☒  $(X_0 \Rightarrow X_1) \wedge (\neg X_2)$

4. Consider the tensor networks:



Which of these formulas is the contraction of the tensor network a) leaving only  $X_0, X_1, X_2, X_3$  open?

☐  $(X_0 \vee (X_1 \Leftrightarrow X_2)) \vee X_3$  ☒  $(X_0 \vee X_1) \Leftrightarrow (X_2 \vee X_3)$  ☐  $Y_2 \Leftrightarrow (Y_0 \Leftrightarrow Y_1)$

Which tensor is the contraction of the tensor network b) leaving only  $Y_2$  open?

☐  $Y_2 \Leftrightarrow (Y_0 \Leftrightarrow Y_1)$  ☐  $\mathbb{I}[Y_2]$  ☒  $\epsilon_1[Y_2]$  ☐  $\epsilon_0[Y_2]$

5. Which of these matrices encodes the constraint that the state of the variable  $X_0$  is by 1 larger than the state of the variable  $X_1$ ?

☐  $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} [X_0, X_1]$  ☒  $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} [X_0, X_1]$

☐  $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} [X_0, X_1]$  ☒  $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} [X_1, X_0]$

6. Let us consider the two propositional formulas

$$f_0[X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] \quad \text{and} \quad f_1[X_0, X_1] = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} [X_0, X_1].$$

Which of these entailment statements is true?

☐  $f_1 \models f_0$  ☐  $\neg f_0 \models f_1$  ☒  $f_1 \models \neg f_0$  ☐  $\neg f_1 \models f_0$  ☒  $f_0 \models \neg f_1$  ☐  $f_0 \models f_1$

7. Which of these tensors is a probability distribution?

☒  $\begin{pmatrix} 0.2 \\ 0.4 \\ 0.4 \end{pmatrix} [X_0]$  ☐  $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} [X_0]$  ☒  $\begin{pmatrix} 0.1 & 0.2 & 0.3 \\ 0.1 & 0.3 & 0 \end{pmatrix} [X_0, X_1]$  ☐  $\begin{pmatrix} 0.1 & 0.2 & 0.3 \\ 0.6 & 0.2 & 0.1 \end{pmatrix} [X_0, X_1]$

8. Let  $\mathbb{P}[X_{[d]}]$  be a distribution of the variables  $X_{[d]}$  and let there be two disjoint subsets  $\mathcal{U}^0, \mathcal{U}^1 \subset [d]$ . Which of these is the distribution of the variables  $X_{\mathcal{U}^0}$  conditioned on  $X_{\mathcal{U}^1}$ ?

☐  $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_{[d]}/\mathcal{U}^0 | X_{\mathcal{U}^1}]}$  ☐  $\langle \mathbb{P}[X_{[d]}] \rangle_{[\emptyset]}$  ☒  $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_{\mathcal{U}^0} | X_{\mathcal{U}^1}]}$  ☐  $\langle \mathbb{P}[X_{[d]}] \rangle_{[X_{\mathcal{U}^0 \cup \mathcal{U}^1}]}$

9. Consider a distribution

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.10 & 0.05 \\ 0.05 & 0.05 \end{pmatrix} \begin{pmatrix} 0.30 & 0.15 \\ 0.15 & 0.15 \end{pmatrix} [X_0, X_1, X_2]$$

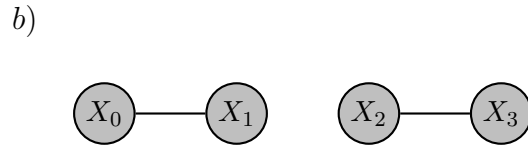
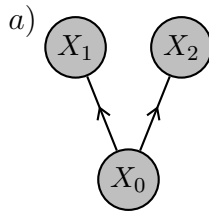
Which of these is the distribution  $\mathbb{P}[X_0, X_1 | X_2]$  of  $X_0, X_1$  conditioned on  $X_2$ ?

- ☐  $\begin{pmatrix} 0.40 & 0.20 \\ 0.20 & 0.20 \end{pmatrix} [X_0, X_1]$
- ☐  $\begin{pmatrix} 0.20 & 0.10 \\ 0.10 & 0.10 \end{pmatrix} [X_0, X_1]$
- ☐  $\begin{pmatrix} 0.10 & 0.05 \\ 0.05 & 0.05 \end{pmatrix} \begin{pmatrix} 0.30 & 0.15 \\ 0.15 & 0.15 \end{pmatrix} [X_0, X_1, X_2]$
- ☒  $\begin{pmatrix} 0.40 & 0.20 \\ 0.20 & 0.20 \end{pmatrix} \begin{pmatrix} 0.40 & 0.20 \\ 0.20 & 0.20 \end{pmatrix} [X_0, X_1, X_2]$

Which of these independence statements is true with respect to this distribution?

- ☒  $(X_0 \perp X_2)$    ☐  $(X_0 \perp X_1) | X_2$    ☒  $(X_1 \perp X_2)$    ☐  $(X_0 \perp X_1)$    ☐  $(X_1 \perp X_0) | X_2$

10. Consider these two hypergraph:



Which of these independence statements is true with respect to any Bayesian network on the hypergraph a)?

- ☐  $(X_0 \perp X_1)$    ☐  $(X_0 \perp X_2)$    ☒  $(X_2 \perp X_1) | X_0$    ☐  $(X_1 \perp X_2)$    ☐  $(X_1 \perp X_2, X_0)$

Which of these independence statements is true with respect to any Markov network on the hypergraph b)?

- ☒  $(X_0 \perp X_2)$    ☐  $(X_0 \perp X_1)$    ☒  $(X_2 \perp X_0) | X_1$    ☐  $(X_2 \perp X_3)$    ☐  $(X_3 \perp X_2)$